

A Tutorial on Graph RAG & Meta KDD Cup

2026.7.10
Data Systems Lab

Overview

- Introduction
 - Retrieval-augmented Generation (RAG)
 - Graph RAG
- Meta KDD Cup
 - Benchmark
 - Baseline Approach

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RAG Mitigates Weakness of Large Language Models

Hallucinations

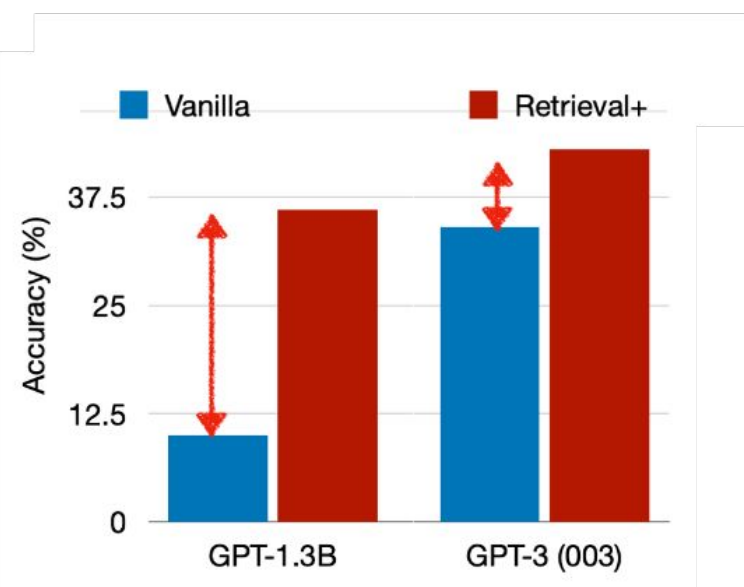
Costs of adaptations

Copyright / privacy

Large parameter size

Significant improvements across model scale, with larger gain with smaller LM

QA



RAG Mitigates Weakness of Large Language Models

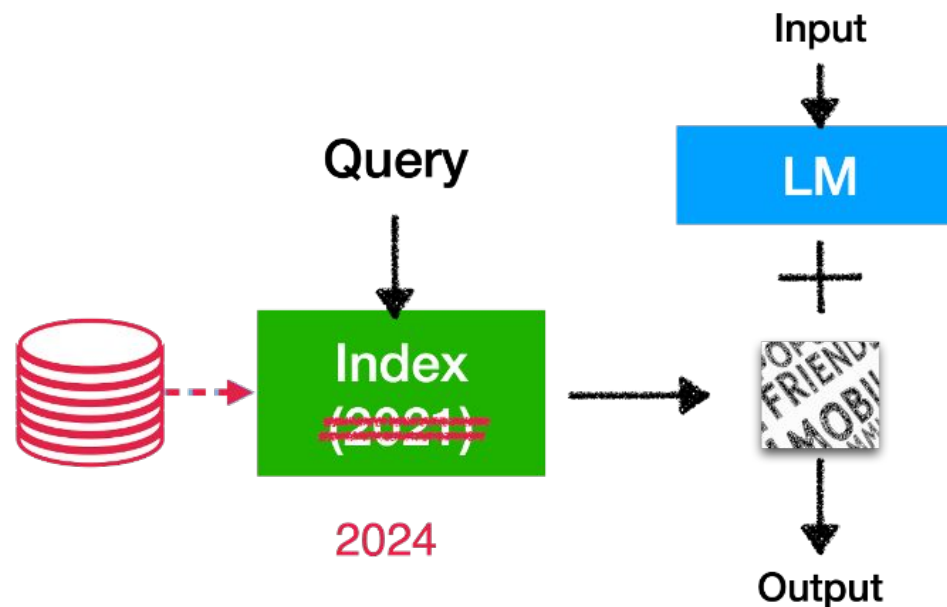
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Replacing Datastore (Index) for adaptations without training



RAG Mitigates Weakness of Large Language Models

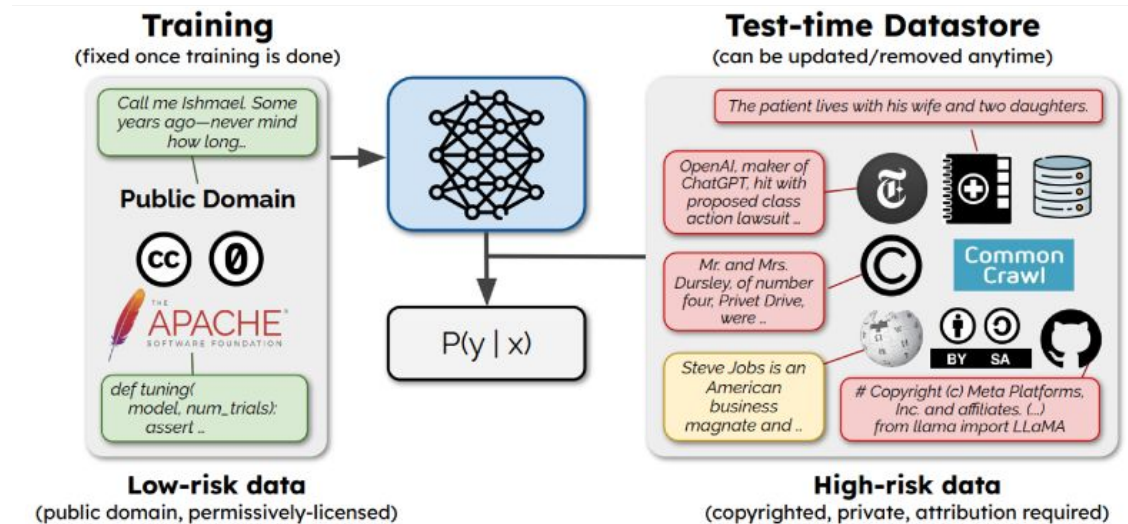
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Segregating copyright-sensitive data from pre-training data



RAG Mitigates Weakness of Large Language Models

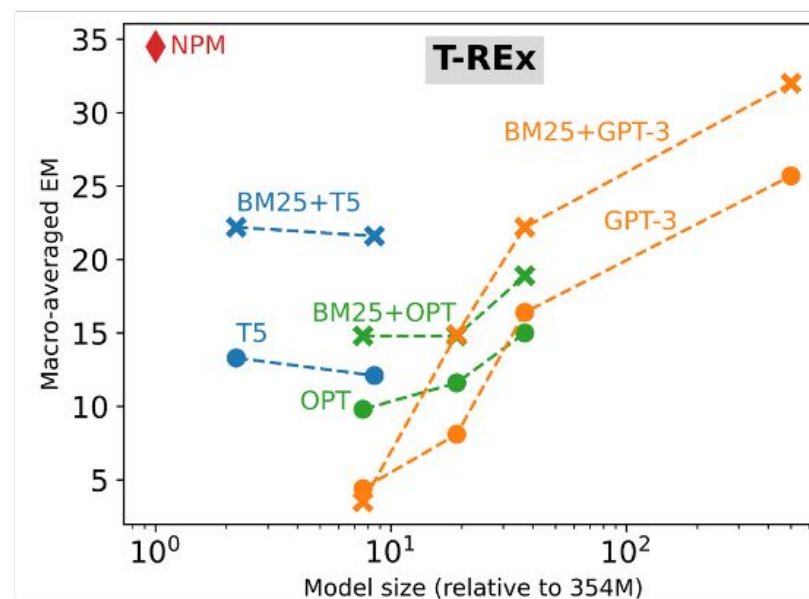
Hallucinations

Costs of adaptations

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Large parameter size

Models with much less parameters can outperforms much larger models!



One of the Hottest Topics in VLDB 2024: (Graph)RAG



Vector Databases

A2 Panel
Vector
Databases:
What's Really
New and
What's Next?

A1
Data
management
and support for
ML/AI

Experimental Analysis of Large-scale Learnable **Vector Storage Compression**
University)*; Penghao Zhao (Peking University); Xupeng Miao (Carnegie Mellon Uni
(Peking University); Bin Cui (Peking University)

SingleStore-V: An Integrated **Vector Database System in SingleStore**
Zhang (Purdue University - West Lafayette); Sasha Podolsky (SingleStore);
Zhou Sun (SingleStore); Robert Walzer (SingleStore); Jianguo Wang (Purdue

Chat2Data: An Interactive Data Analysis System with RAG, **Vector Databases and LLMs** xi
Guoliang Li (Tsinghua University)*

Graph Databases



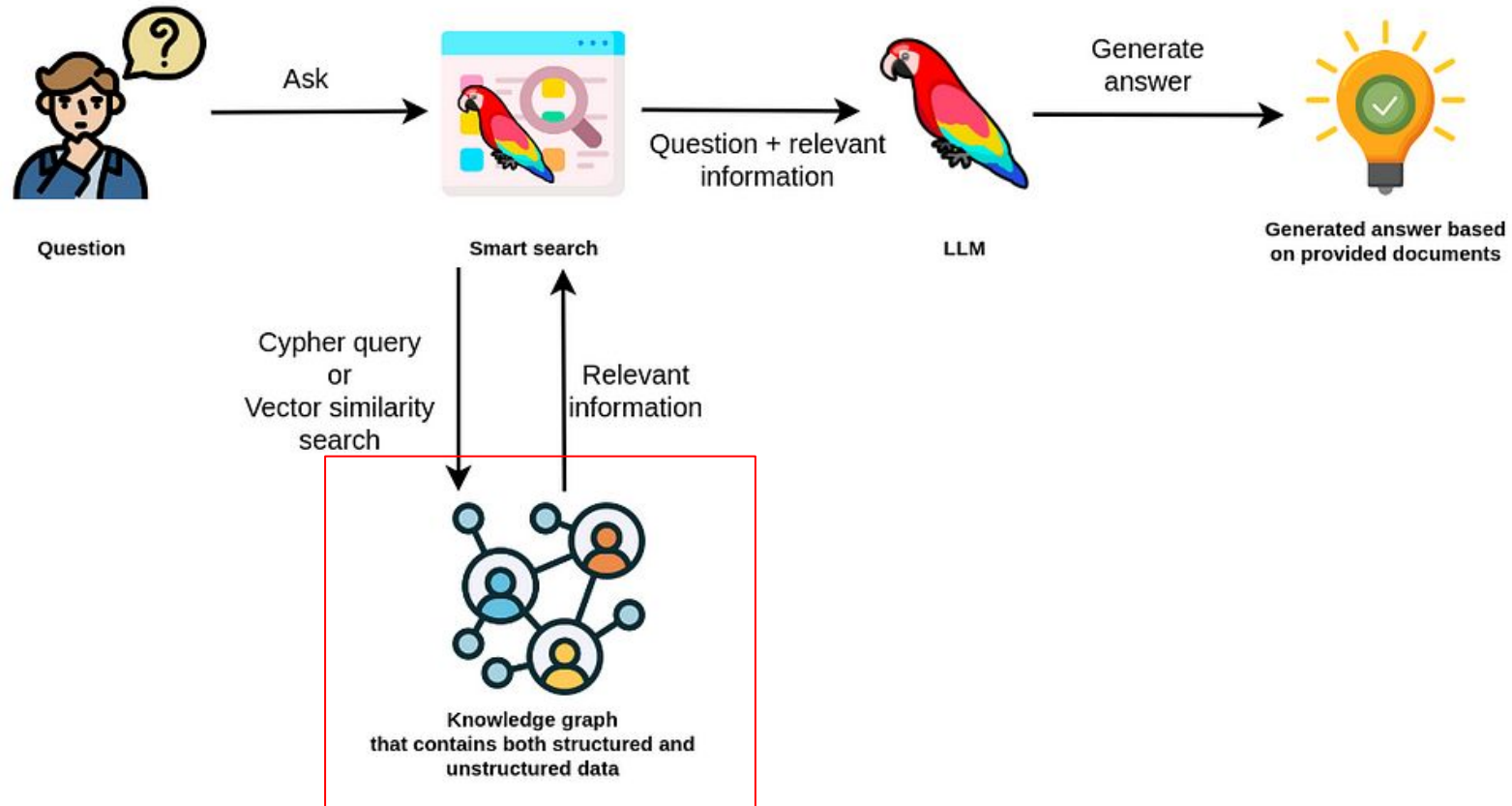
Industry Talk: Integrating GenAI with Graph: Innovations and Insights from
NebulaGraph

Siwei Gu & Yihang Yu (NebulaGraph, China)

LLM+KG
Workshop



GraphRAG Overview



GraphRAG Overview (From Neo4j Document)

Overview

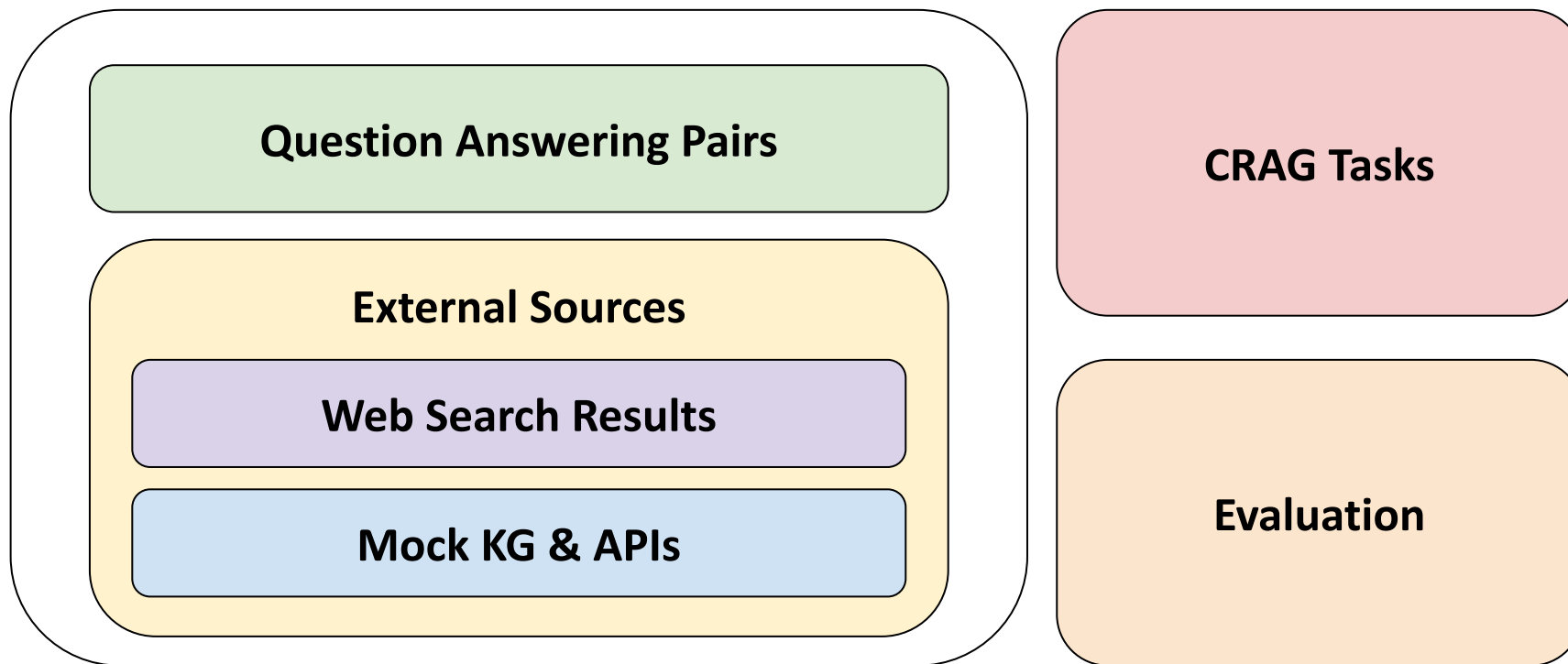
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Meta KDD Cup 2024

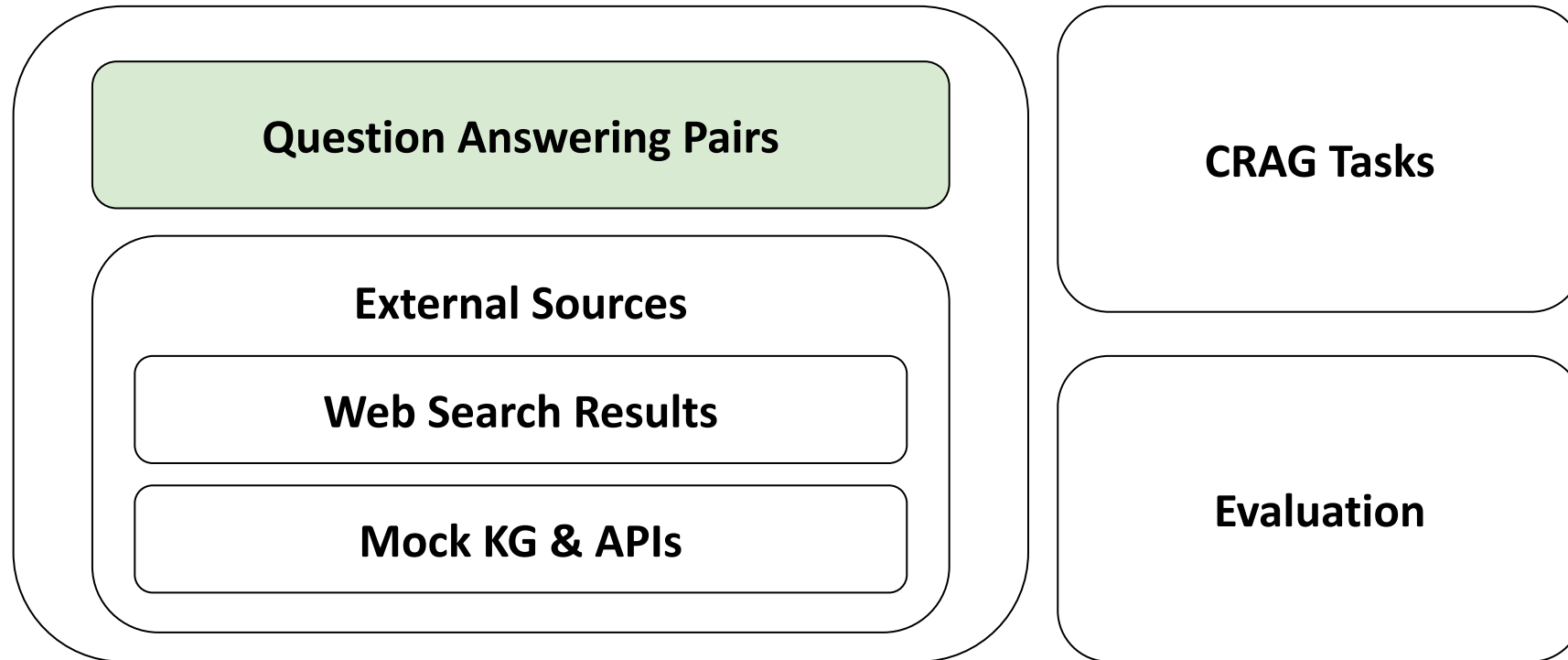
- 2300+ participants, 384 teams, 5600+ submissions
- CRAG benchmark was listed in HuggingFace "Daily papers".  Hugging Face
- Our team was the only Korean team to receive an award, achieving First Place in the Comparison Question category for Tasks 1, 2, and 3!!



CRAG Benchmark Overview



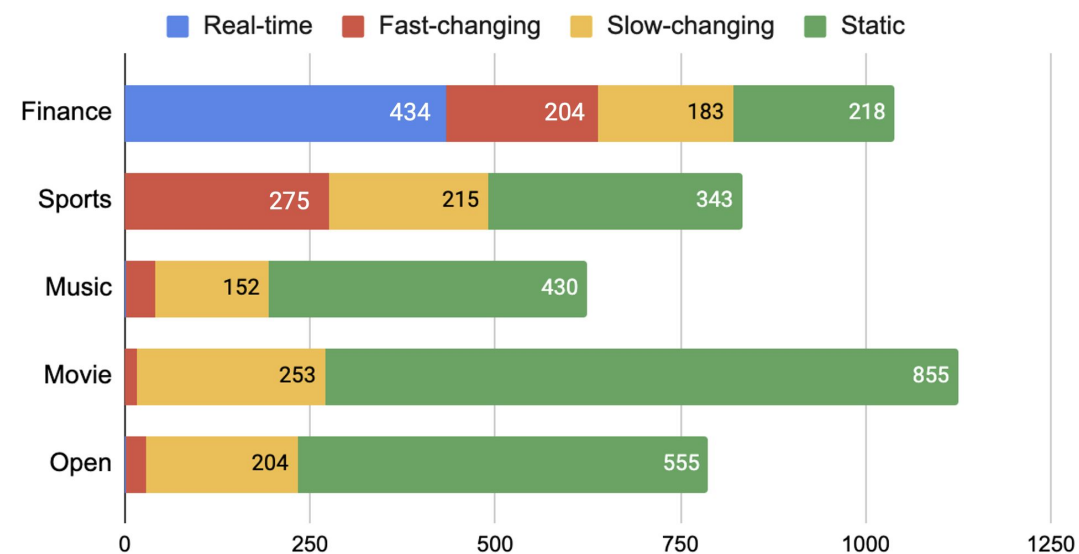
CRAG Benchmark Overview



Question Answering Pairs

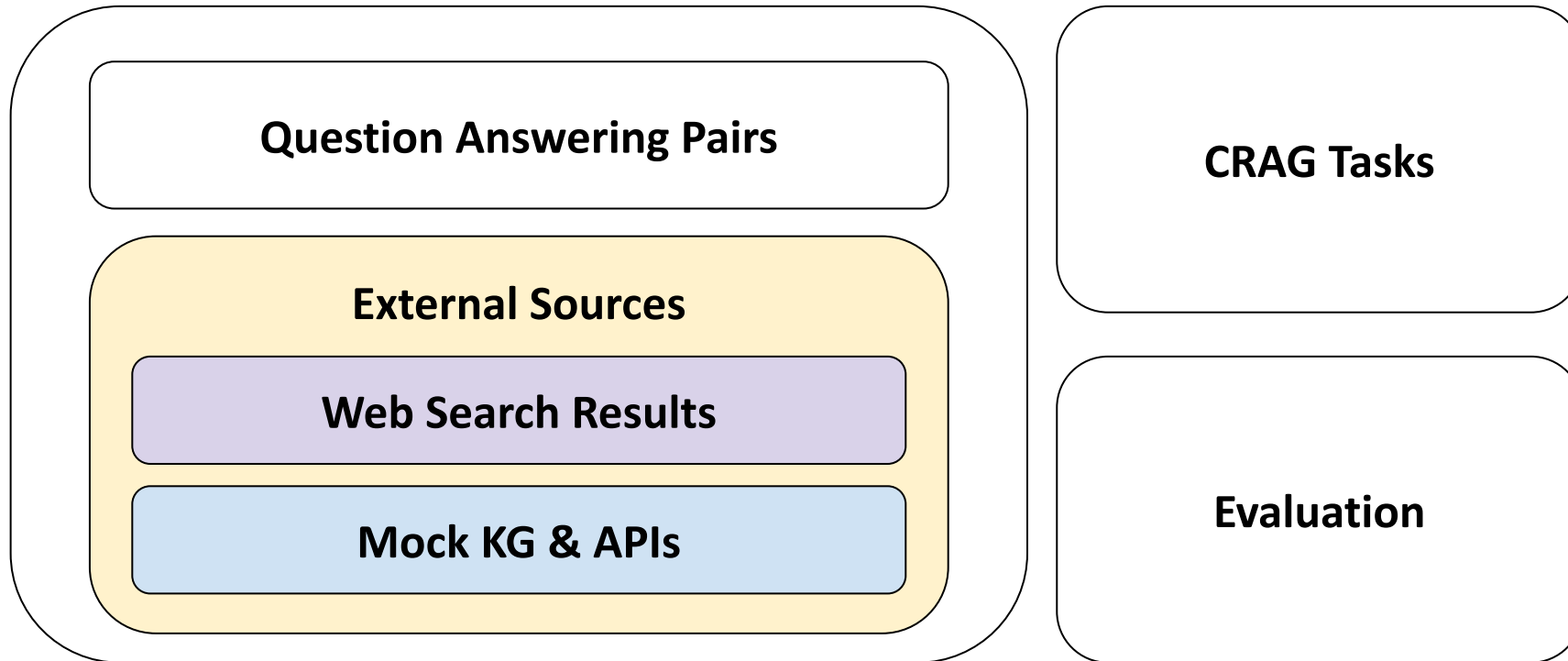
- 4400+ QA pairs from 5 domains (Finance, Sports, Music, Movie, Encyclopedia)
- Questions for static, slow-changing, fast-changing, and real-time information
- Questions for head, torso, and tail entities
- Simple-fact questions and complex questions

Real-time, Fast-changing, Slow-changing and Static



Total	Simple	Simple w. Cond	Set	Comparison	Aggregation	Multi-hop	Post-Processing	False Premise
4409	1205	689	403	546	489	382	180	525

CRAG Benchmark Overview



External Sources

- Web Search Results: 50 webpages for each question from BraveAPI web search
- Mock KG & APIs
 - Mock KG: 2.6M entities
 - Mock APIs: 38 mock APIs

Q1:

What's the latest film that walt becker has directed?

Q2:

Which one of these came out earlier, the greater meaning of water or small town ecstasy?

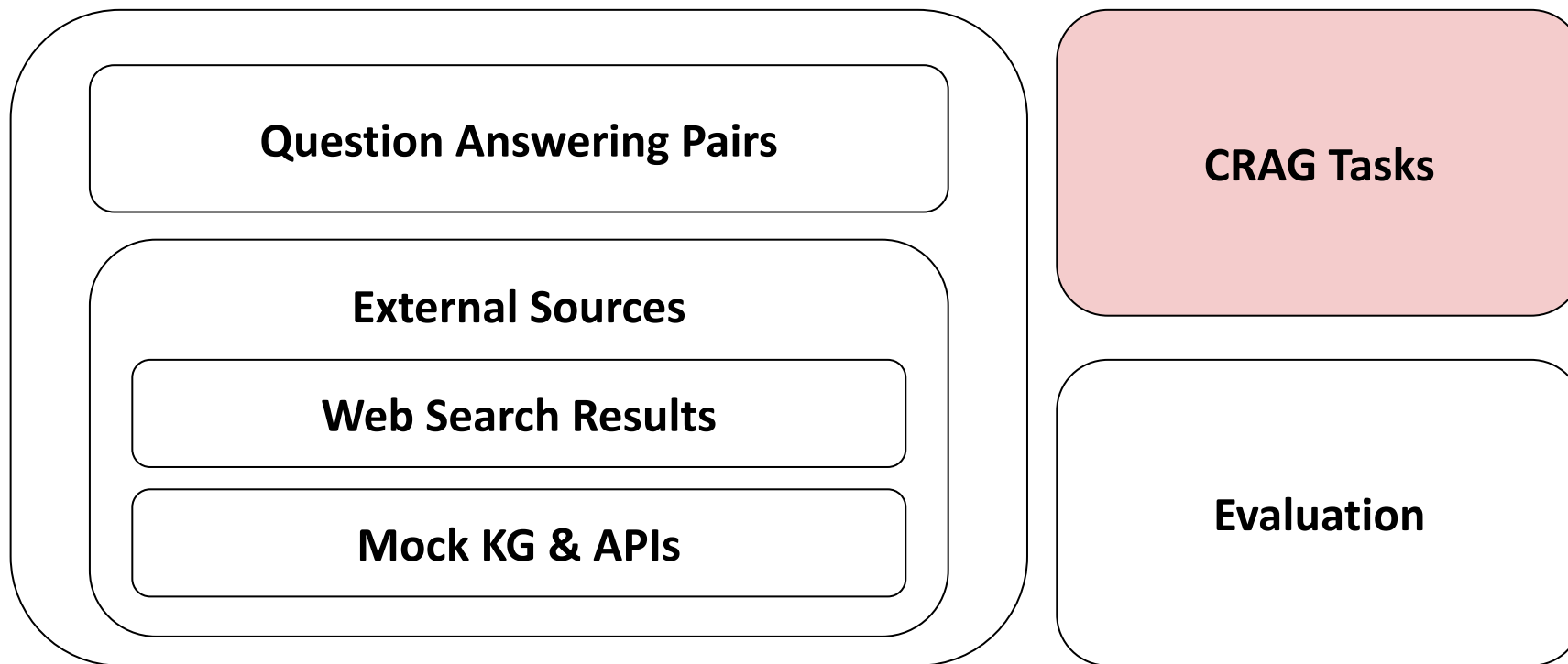
API for Q1:

```
get_movie_person_crew(None, "walt becker", eq(job, "Director"));
sort(None, -year)["movie_name"]
```

API for Q2:

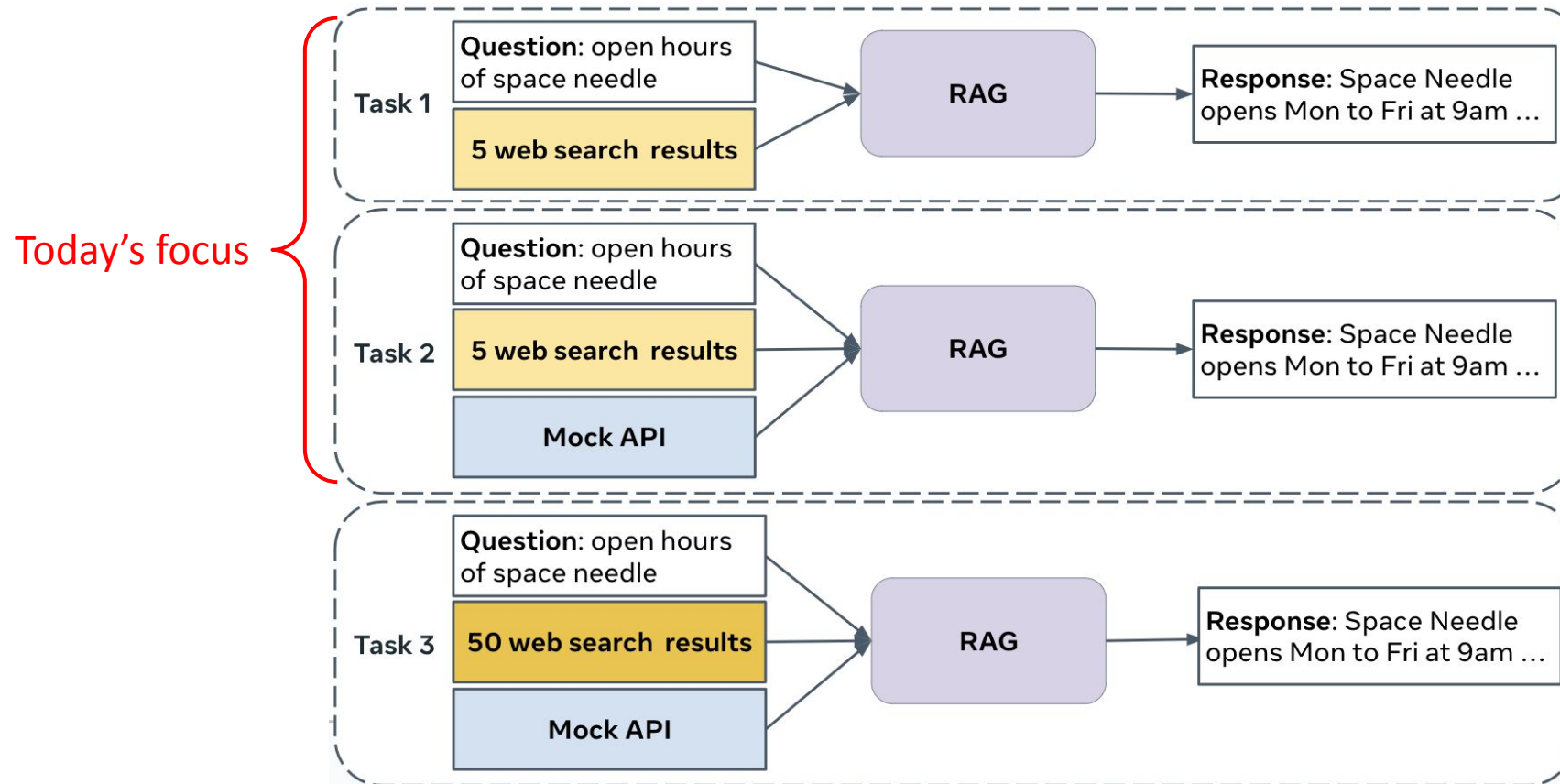
```
get_movie("greater meaning of water")["release_date"];
get_movie("small town ecstasy")["release_date"]
```

CRAG Benchmark Overview

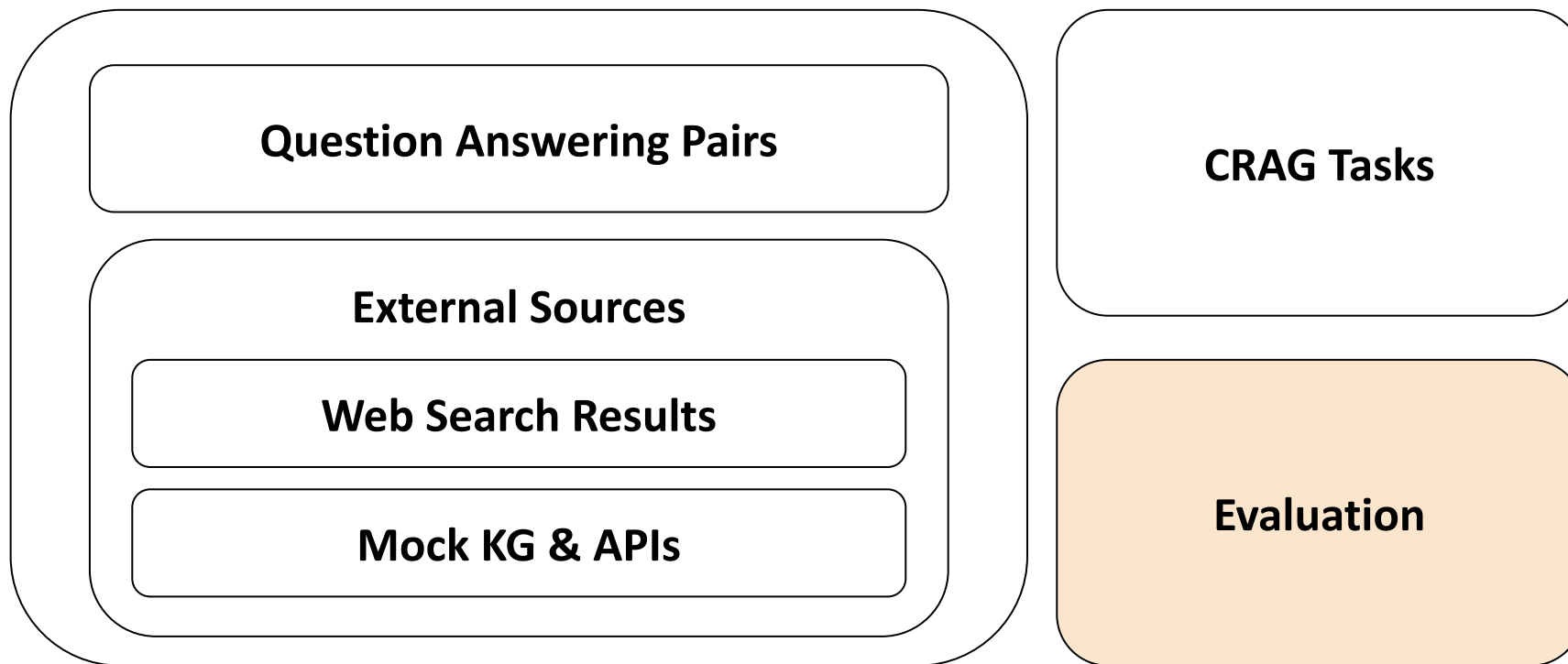


CRAG Tasks

Three tasks build up information gradually to test different capabilities of RAG systems.



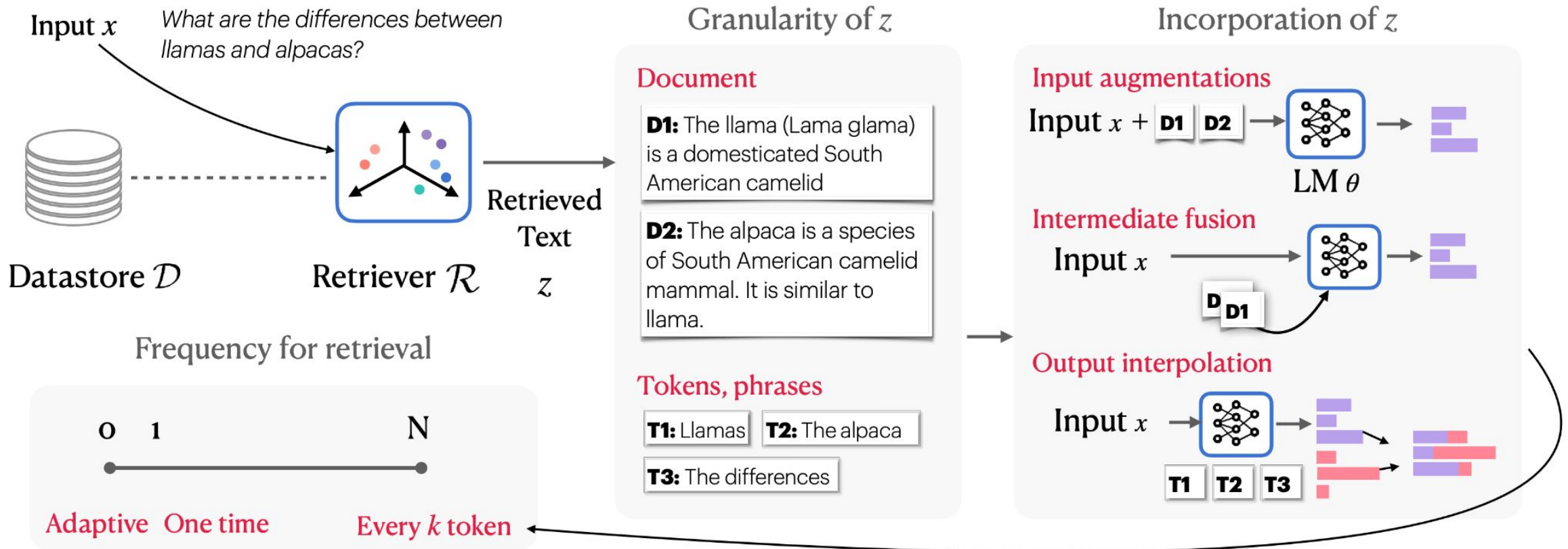
CRAG Benchmark Overview



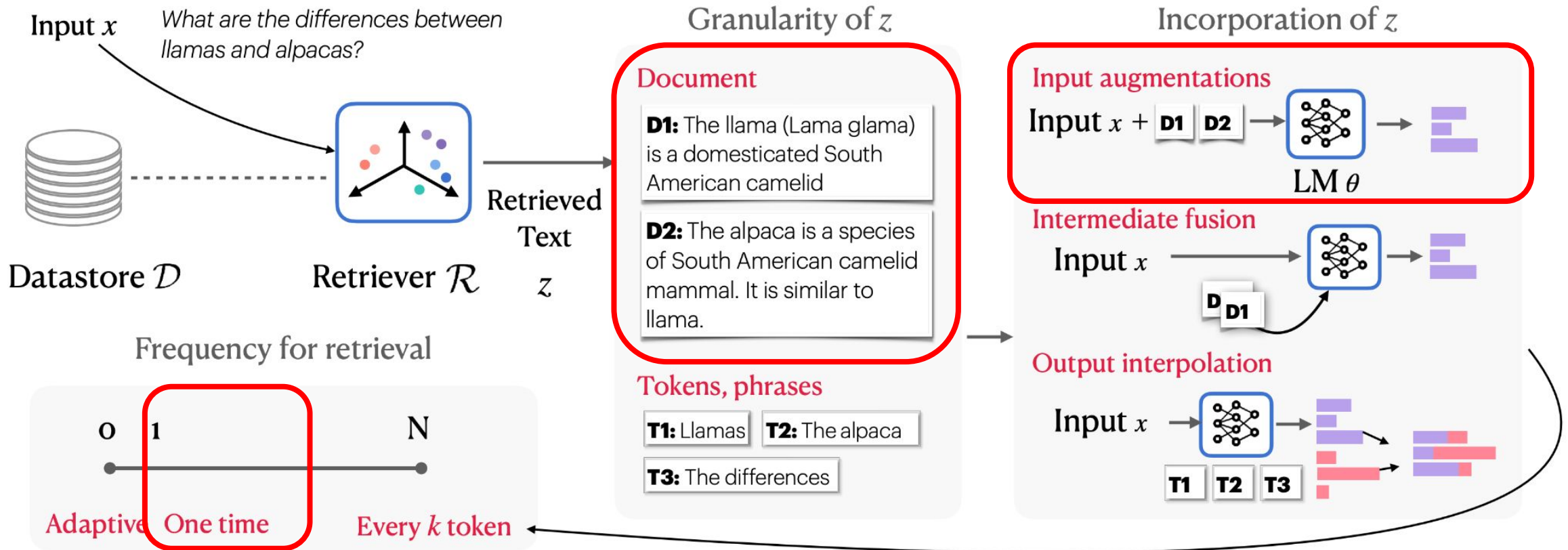
Evaluation

- Metrics: $\text{CRAG Score} = \text{Exact Accuracy} + 0.5 * \text{Accuracy} - \text{Hallucination rate}$
 - **Exact Accuracy:** The percentage of questions for which the generated answer exactly matches the ground truth answer.
 - **Accuracy:** The percentage of questions for which the generated answer is not exact but has the same meaning as the ground truth.
 - **Hallucination:** The percentage of questions for which an incorrect answer was generated.
 - **Missing:** The percentage of questions where the response was "I don't know."
- Evaluation
 - Auto-eval (with GPT-4)
 - Manual-eval (with Human)

Diverse Architectures of RAG

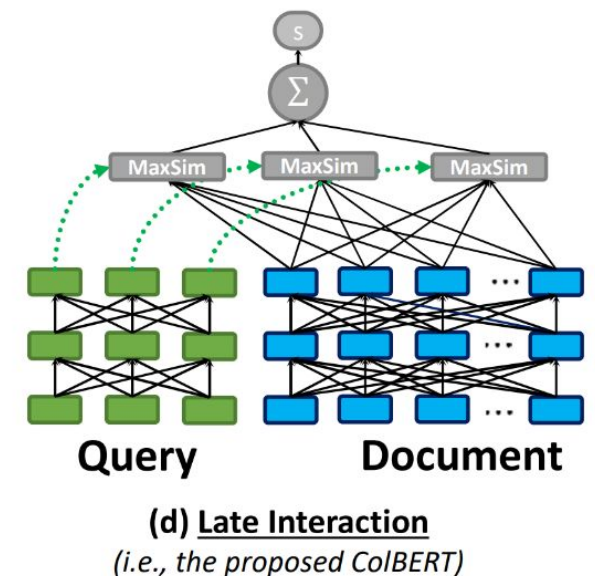
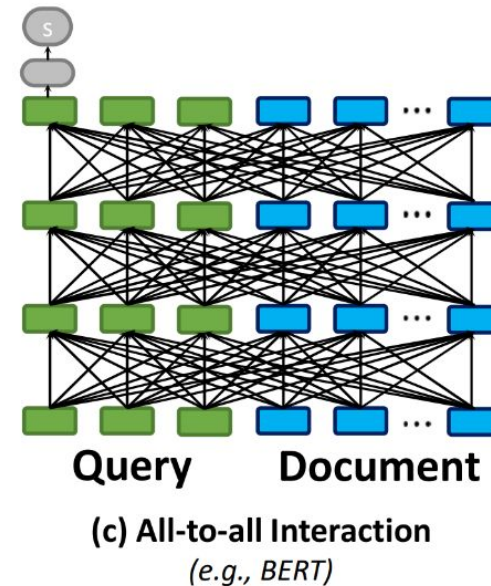
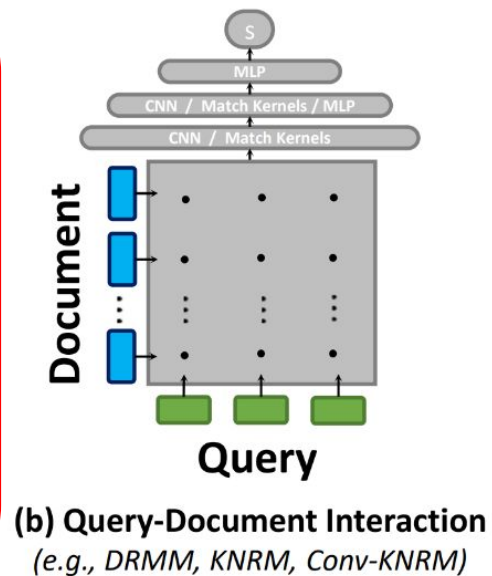
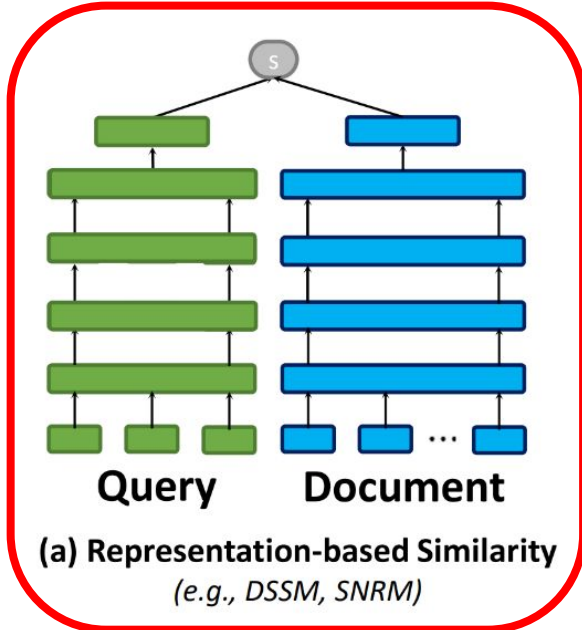


Baseline Approach



Retrieval Module for Web Search Results

- Retriever performs representation-based similarity search.
- It retrieves top-k relevant sentences.



Mock APIs & Mock KG

- Task 2 provides mock APIs to query the provided mock knowledge graph (mock KG).

Q1:

What's the latest film that walt becker has directed?

Q2:

Which one of these came out earlier, the greater meaning of water or small town ecstasy?

API for Q1:

```
get_movie_person_crew(None, "walt becker", eq(job, "Director"));
sort(None, -year)["movie_name"]
```

API for Q2:

```
get_movie("greater meaning of water")["release_date"];
get_movie("small town ecstasy")["release_date"]
```

- The mock KG, as a structured knowledge base, offers precise information; however, generating an accurate query is essential for retrieving correct answers.

Retrieval Module for Mock KG

- The knowledge graph retrieval module follows these three steps:
 - a. The LLM generates the query domain and API arguments.
 - b. Based on the generated query domain and API arguments, a decision tree is used to sequentially call the appropriate mock APIs.
 - c. The results from these mock API calls are provided to the LLM along with retrieved results from web search results.
- We aim to go a step further by leveraging the Model Context Protocol (MCP) to upgrade the knowledge graph retrieval module.

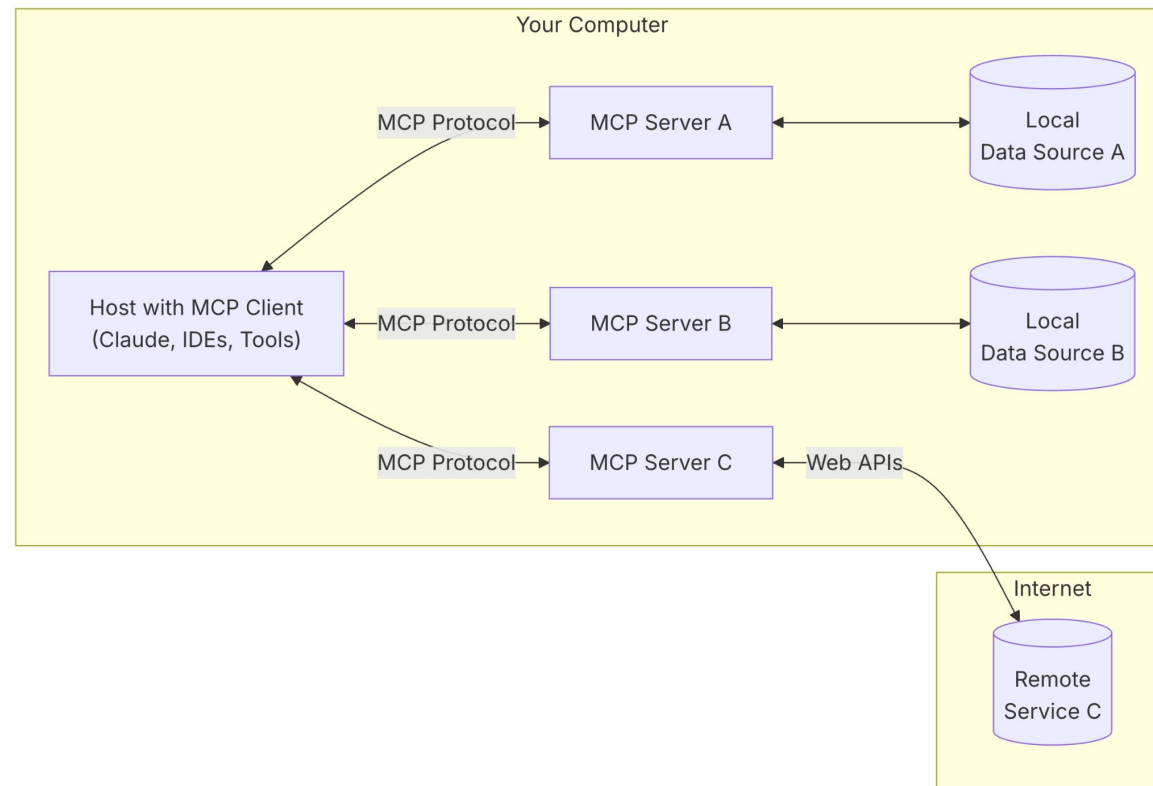
Model Context Protocol

Model Context Protocol (MCP) is an open protocol that standardizes how applications provide context to LLMs.

- Think of MCP like a USB-C port for AI applications.
- Just as USB-C provides a standardized way to connect your devices to various peripherals and accessories, MCP provides a standardized way to connect AI models to different data sources and tools.

General Architecture

At its core, MCP follows a client-server architecture where a host application can connect to multiple servers:



Appendix